

PRAMANA

Evidence-Centric Attribution Platform for Dark Web Threat Actors

Technical Project Dossier | SIH Problem Statement 26151 | NTRO | Blockchain & Cybersecurity
Agency-style engineering and explanatory document for technical, academic, and evaluation audiences

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CENTRAL THESIS

The project is not primarily about finding more traces. It is about deciding what a trace is worth once evidence may be duplicated, correlated, common, planted, time-dependent, or later retracted.

This dossier is written so that a first-time reader can follow the system from basic concepts to the mathematical and architectural details, while a technically advanced reader can inspect the assumptions, controls, limits, evaluation design, and failure modes.

Important status statement: the project record describes the research audit, architecture, specification, and presentation work as complete; implementation and benchmark measurement are the next phase. No production accuracy, deployment history, or government endorsement is claimed in the source material.

Source basis: Primary project artefacts supplied in this conversation: PRAMANA_Master_Blueprint.pdf, PRAMANA_Master_Narrative.pdf, and the PRAMANA presentation deck.

Document Control and Reading Guide

Field	Value
Project	PRAMANA - Evidence-Centric Attribution Platform for Dark Web Threat Actors
Problem Statement	Smart India Hackathon 2026 - Problem Statement 26151
Organisation / Theme	NTRO / Blockchain & Cybersecurity
Purpose	Technical explanation, design interpretation, architecture, methodology, evaluation, and implementation guide
Primary source hierarchy	Master Blueprint > supporting narrative > presentation content where they differ
Current status in source record	Designed and specified; presentation built; implementation not yet begun; no benchmark results yet
Evidence discipline	Measured result vs design target vs illustrative case vs future capability must remain explicitly separated

How to Read This Dossier

1. Read Sections 1-4 to understand the problem, the reframing, and the system boundaries before reading the mathematics.
2. Read Sections 5-9 to understand the evidence model, entity resolution, fusion, counter-evidence, time, and provenance.
3. Read Sections 10-14 to understand the architecture, intelligence modules, AI/ML boundaries, analyst workflow, and reporting.
4. Read Sections 15-18 to understand evaluation, test environments, security and adversarial resilience, legal/forensic posture, and current implementation status.
5. Use Appendices A-D as quick-reference material during a presentation or technical viva.

NOTATION USED THROUGHOUT

A/B or H1/H2 denote competing identities or hypotheses. LR means likelihood ratio. $\log_{10}$ LR is the scoring space used by the design. k denotes the count of independent evidence families supporting an assessment. “EXCLUDED” is a hard state, not a low score.

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- 14. Analyst Workflow and Product Experience** - Investigation workspace, balance sheet, review gate, provenance, and reports.
- 15. Evaluation Science and Test Harness** - Cllr, Tippett plots, calibration, clustering metrics, ablations, and leakage controls.
- 16. Controlled Test Environments** - RANGE-SIM and RANGE-TOR, their purpose, and what their metrics can and cannot prove.
- 17. Threat Model, Adversarial Resilience, and Failure Modes** - What an adversary can do, what PRAMANA catches, and what remains fundamentally hard.
- 18. Legal, Forensic, Ethical, and Governance Posture** - Scope, evidence handling, data protection, and the limits of legal claims.
- 19. Implementation Status and Roadmap** - What is complete, what is specified, and what comes next.
- 20. Strategic Assessment** - The project's strongest contribution, risks, and future evolution.
- Appendix A. One-Page Technical Summary** - Fast reference for viva / judging.
- Appendix B. Glossary** - Plain-English and technical definitions.
- Appendix C. Judge / Viva Questions and Defensible Answers** - High-probability questions and exact framing.
- Appendix D. Claim Discipline Checklist** - What can be said without overclaiming.

1. Executive Understanding

PRAMANA is a proposed investigative-support platform for linking pseudonymous dark-web identities using multiple weak and heterogeneous signals while explicitly controlling for the ways those signals become misleading. The platform is designed around a simple proposition: an attribution system should not merely report that two things look similar; it should show how much the observed evidence supports a stated hypothesis, what has been discounted, what argues against it, and how the computation can be reconstructed later.

The project is therefore better understood as an evidence-accounting system than as a scraper, a graph visualizer, a stylometry product, or an autonomous AI investigator. Collection supplies observations. The evidence ledger determines what those observations are worth. The intelligence engine generates and tests hypotheses. The analyst remains the author of the conclusion.

THE THREE-LINE MENTAL MODEL
The graph is a view. The evidence ledger is the truth. The human is the author of the conclusion.

What PRAMANA is trying to change

Conventional failure	PRAMANA response
Five copies of one artefact look like five votes.	Canonicalise to one artefact; group by independence origin; damp within-family corroboration.
Common indicators are treated as highly informative.	Compute rarity against a declared reference population; suppress hubs.
A single similarity score implies confidence.	Return a likelihood-ratio assessment with a verbal band, evidence-family count, and defence hypothesis.
Contradictions are noticed manually.	Run an explicit counter-evidence pass; subtract soft contradictions; hard-veto to EXCLUDED.
Graph edges become facts.	Store evidence in an append-only bitemporal ledger; generate graph views from the ledger.
Machine output silently becomes an analyst conclusion.	Enforce a human review gate, rationale requirement, and provenance-backed assessment.

What success means

The project does not define success as “the system names a real person.” Success is defined more narrowly and more defensibly: for a proposed link between two pseudonymous identities, the system should produce a calibrated, traceable and challengeable assessment, or explicitly refuse to produce one when its safeguards are not satisfied.

What the current project record actually proves

- The research was instrumented, audited, and converted into a detailed architecture and execution plan.
- The core scoring design, data model, interface, evaluation protocol, and controlled test environments are specified.

- The presentation layer and demonstration logic are specified in detail.
- Implementation and benchmark results are not yet claimed in the supplied project record.

Source basis: Blueprint opening thesis; Narrative executive overview; Blueprint final red-team and status sections.

2. The Problem: Attribution as Evidence Arithmetic

A dark-web operator does not appear to investigators as a single clean identifier. The same underlying actor may appear through several marketplace accounts, forum handles, PGP keys, payment addresses, files, product images, infrastructure traces, contact identifiers, and temporal behaviour. Conversely, the same surface may be shared by many unrelated actors. This creates a many-to-many inference problem.

The dangerous shortcut is to treat every observed similarity as an independent vote. That can produce a false merge: one shared escrow address, one default PGP block, one copied template, one common hosting provider, and one forum writing style may all appear to corroborate the same person even though their common cause is simply the surrounding ecosystem.

The asymmetry of error

A missed link usually delays an investigation. A false link can contaminate later hypotheses because downstream analysts may reuse the merge as an input. PRAMANA therefore deliberately biases toward refusal, constraints, reversibility, and transparent uncertainty rather than maximum linkage recall.

CORE RISK

More data can make a weakly designed system more confidently wrong. Volume is not a defence; evidence weighting and independence accounting are.

The problem in formal terms

Given two candidate identities A and B, evaluate H_{same} versus H_{diff} .
 Observe heterogeneous evidence $E = \{e_1, e_2, \dots\}$.
 Determine which observations are genuinely distinct, how rare each indicator is, which evidence families they belong to, what contradicts the hypothesis, and whether the remaining support is sufficient to produce a defensible assessment.

The difficulty is not simply classification. It is inference under correlated, adversarial, time-dependent, partially contaminated observations with asymmetric error costs.

3. Definitions and Conceptual Model

A technically serious system fails if its data concepts are ambiguous. PRAMANA therefore separates terms that are often collapsed in ordinary graph tools.

Term	Definition	Why it matters
Observation	A captured fact or artefact occurrence, before it is interpreted as evidence for a hypothesis.	Prevents raw data from being treated as a conclusion.
Evidence	An observation judged relevant to a	Makes evidential value contextual

Term	Definition	Why it matters
	specific investigative question and assigned a family, rarity, validity, and provenance context.	rather than intrinsic.
Canonical artefact	The deduplicated representation of materially identical or near-identical captured content.	Prevents mirrors and reposts from becoming false corroboration.
Evidence group	A set of evidence items sharing an independence origin within a family.	Allows the system to discount correlated observations.
Entity	A represented thing such as an account, key, wallet, service, or other object that exists in the data model.	Separates things from hypotheses about who controls them.
Relationship	A typed, weighted, time-bounded relationship between entities.	Makes edges explainable and revisable.
Hypothesis	A named proposition, such as “A and B are controlled by the same persona.”	Separates inference from observation.
Assessment	A structured result containing LR, verbal band, k, counter-evidence, defence hypothesis, provenance, and review state.	Replaces a single opaque score.
Reference population	The explicit population against which indicator rarity is estimated.	Makes LR-like weights interpretable and reproducible.
Bitemporal record	A record that tracks when an event is valid and when the system believed/revised the fact.	Enables historical reconstruction and retraction propagation.

The evidence hierarchy

Source -> Capture -> Seal -> Canonicalise -> Extract -> Group -> Weigh
-> Hypothesis -> Fuse -> Assessment -> Human Review

Each layer is deliberately restricted. A hypothesis cannot become an evidence source; an entity cannot be created merely because a fusion score is high; a relationship cannot exist without a trail back to the underlying evidence.

4. Design Philosophy and Non-Negotiable Boundaries

PRAMANA treats boundaries as engineering inputs, not disclaimers. The system is defined as lawful, defensive, authorised, and investigative-support oriented. That scope deliberately excludes capabilities that would require attacking Tor, accessing credentialed criminal marketplaces, bypassing access controls, acquiring inherently unlawful content, or automatically identifying private individuals.

The project supports	The project explicitly does not claim
Lawful evidence discovery and archival capture	De-anonymisation of the Tor network or defeat of Tor

The project supports	The project explicitly does not claim
	cryptography
Cross-source and cross-marketplace correlation	Automatic real-world identity production
Probabilistic entity resolution at pseudonym level	Deterministic Monero or modern privacy-coin tracing to evidentiary standard
Infrastructure candidate analysis on authorised/self-owned targets	Passive enumeration of all hidden services
Stylometric corroboration with hard caps	Infallible stylometric identification
Human-reviewed attribution hypotheses	Replacement of analyst judgement
Evidence-backed reports and export bundles	Automatic court admissibility or government endorsement

Why the boundaries improve the method

Because the collection layer is restricted to lawful and reproducible strata, the project cannot rely on privileged access to build a perfect view of the hidden ecosystem. The architecture therefore has to be good at extracting value from noisy, partial and adversarial evidence. The constraints are the reason the project emphasizes canonicalisation, rarity, independence, counter-evidence, provenance, and refusal.

Source basis: Narrative constraints page; Blueprint problem-statement reinterpretation and refusal list.

5. End-to-End Evidence Lifecycle

The system can be understood as a chain of increasingly stronger claims. Each transition is a controlled transformation with a defined failure mode.

1. Source

A lawful source is identified and assigned source metadata, legal basis, reliability grade, and independence context.

2. Capture

The source material is captured using an archival mechanism appropriate to the collection mode.

3. Seal

The capture is hashed and linked into a provenance chain; timestamping support is retained.

4. Canonicalise

Duplicate or near-duplicate content is collapsed into a canonical artefact identity.

5. Extract

Deterministic parsers and validators extract identifiers, metadata, structural features, and signals.

6. Group

Evidence is assigned independence keys so observations with the same underlying origin do not stack linearly.

7. Weigh

Rarity, source quality, temporal validity, family rules, hub status, and other controls determine contribution.

8. Hypothesis

The system states exactly what proposition is being evaluated.

9. Fuse

Evidence is combined using capped, damped log-LR logic after counter-evidence processing.

10. Assess

The system attaches a verbal band, k, provenance, limitations, and the competing defence hypothesis.

11. Human review

The analyst sees evidence, then counter-evidence, then the score, and records the decision.

WHY THE ORDERING MATTERS

If the system scores before it knows whether observations are duplicate, common, or contradicted, then the final number can be mathematically tidy but evidentially meaningless.

6. Canonicalisation, Independence, and Rarity

6.1 Canonicalisation

Canonicalisation answers the question: “Are these observations materially the same artefact?” For images the design uses perceptual hashes such as pHash. For text and documents it can use SimHash and structural fingerprints. Listing templates are represented through section order, table structure, signature-block form, units, and price-rounding conventions.

23 raw observations

|

+-- mirrored / quoted / archived copies

v

1 canonical artefact

v

1 evidence group

v

1 independent contribution

6.2 Independence keys

Independence asks whether two observations represent separate opportunities to observe the underlying proposition. A document mirrored to five sites may generate five retrieval events, but it is not five independent facts. Evidence with a common independence key is grouped before fusion.

6.3 Within-family damping

After grouping, the strongest evidence group in a family contributes fully and subsequent independent groups receive a damping factor. The initial project parameter is $\lambda = 0.2$. This means the second independent group within the same family contributes one fifth of its raw strength, the third contributes one twenty-fifth, and so on, subject to the family cap.

6.4 Rarity and hub suppression

PRAMANA uses IDF-like rarity over a declared reference population. An indicator that appears in almost everyone carries little information. In addition, high-frequency indicators are treated as hubs. The current design threshold is $\tau = 12$ corpus occurrences; above that threshold the indicator can be flagged as a hub and its evidential contribution forced to zero.

Indicator example	Interpretation	PRAMANA treatment
Rare PGP fingerprint	Potentially highly informative	Can contribute strongly if provenance and key-sharing risks are controlled.
Self-hosted rare domain	Potentially informative infrastructure signal	Weight by rarity and apply shared-hosting counter-check.
Common email provider domain	Very common	Low or zero information.
Marketplace default PGP block	Shared hub	Suppress to zero and show the reason.
Same product image	Useful artefact continuity	Canonicalise and group across mirrors before weighting.

7. Entity Resolution and Constraint-Aware Clustering

Entity resolution is the step that turns candidate relationships into tentative identity clusters. PRAMANA uses the Fellegi-Sunter probabilistic record-linkage tradition, with Splink as a reference implementation. The key difference is that probabilistic pair scores do not automatically become permanent identity merges.

Candidate generation

- > blocking on multi-key identifiers + MinHash LSH
- > field comparison levels
- > m/u estimation by EM
- > pairwise evidence score
- > must-link / must-not-link constraints
- > capped clustering
- > component-size and hub checks
- > measured cluster quality

Why no naive transitive closure?

Suppose A links to B and B links to C. It does not follow that A links to C. Each pair may be supported by different evidence, and a shared hub may have created both edges. PRAMANA therefore does not let transitive closure silently manufacture an identity. Component-size caps, must-not-link constraints, hub removal, and merge provenance are part of the design.

Must-link versus must-not-link

Must-link expresses evidence that strongly requires two records to stay together. Must-not-link expresses an incompatibility that prevents a merge. A hard must-not-link can survive otherwise impressive similarity because it is a logical constraint rather than another weak vote.

Primary ER metrics

- B-cubed precision/recall for cluster-level correctness.
- Variation of Information for over- versus under-merging.
- False-merge rate as the headline safety-oriented metric.
- Giant-component checks to detect pathological collapse.

8. Evidence Fusion, Likelihood Ratios, and Confidence

PRAMANA uses likelihood-ratio reasoning because the meaningful question is not “How similar are these records?” but “How expected is this observed evidence under the same-actor proposition compared with the different-actor proposition?” The design uses log10 LR as its working scale.

8.1 Why log10 LR?

Likelihood ratios multiply naturally across independent evidence. Taking a base-10 logarithm converts multiplication into addition, which makes a ledger of contributions easier to inspect. PRAMANA does not, however, assume independence blindly. Independence grouping and within-family damping are inserted before the addition.

```
For each family F:
  group = collapse(evidence, independence_key)
  sort groups by strength
  family_score = sum(w_i * lambda^i)
  family_score = min(family_score, CAP[F])

total = sum(family_score) - counter_evidence
require independent_families >= 2
total = min(total, 4.0)
if hard must-not-link: return EXCLUDED
```

8.2 Evidence-family caps

Family	Examples	Initial cap (log10 LR)	Design reason
F1 Cryptographic	PGP, signatures, key metadata, SSH host keys	4.0	Strong, but keys may be shared, stolen, or

Family	Examples	Initial cap (log10 LR)	Design reason
			transferred.
F2 Financial	Address reuse, co-spend heuristics, payment cadence	2.0	CoinJoin, PayJoin, exchange batching and labels reduce certainty.
F3 Infrastructure	TLS/SAN, favicon, headers, clock skew, hosting	2.5	CDNs and shared hosting create common signals.
F4 Contact	Messaging IDs, email, migration notices	3.0	Often rare, but self-declared identities can be forged.
F5 Content artefacts	Image pHash, templates, EXIF, formatting	2.5	Robust to prose laundering, but templates can also be copied.
F6 Linguistic	Stylometry, character n-grams, style embeddings	1.0	Fragile under translation, imitation, topic confounds and LLM rewriting.
F7 Behavioural	Migration co-timing, lifecycle behaviour	0.7	Useful but coarse and vulnerable to automation or staff rotation.
F8 Social	Vouching, co-membership, reciprocal links	1.0	Can be manufactured by sybils or social engineering.
F9 External	Authorised legal-process returns, official notices, credible public corroboration	3.0	Potentially high value, but provenance and source credibility still matter.

8.3 Verbal bands

log10 LR	PRAMANA band
0-1	LIMITED
1-2	MODERATE
2-3	MODERATELY STRONG
3-4	STRONG

The project deliberately caps the global output at 4.0 log10 LR and refuses to emit a real-world identity LR. The purpose of the ceiling is to prevent unbounded aggregation from implying certainty beyond what the architecture is willing to defend.

9. Counter-Evidence, Exclusion, and Self-Challenge

A system that only accumulates support will eventually find a reason to agree with itself. PRAMANA therefore makes counter-evidence a first-class stage rather than a manual analyst afterthought.

Soft contradiction

A soft contradiction reduces the balance. Examples include an ordinary timing inconsistency, style divergence at a suspicious point in the timeline, or other evidence that weakens but does not logically destroy the hypothesis.

Hard contradiction

A hard contradiction is treated as a veto. The output becomes EXCLUDED and the system emits no numerical score. This is intentionally stronger than “a very low confidence score,” because a low score still invites a stressed analyst to bargain with the number.

```
supporting evidence -> candidate score
counter-evidence pass
|
+-- hard contradiction? --> EXCLUDED (no number)
|
+-- otherwise subtract soft contradiction
v
final fusion + review gates
```

Five recurring contradiction classes

- Temporal impossibility or incompatible event ordering.
- Contradictory identifiers or ownership evidence.
- Geolocation or language conflict beyond plausible variation.
- Known impersonation / compromise patterns.
- Evidence that the apparent link is explained by a shared ecosystem rather than shared control.

DESIGN PRINCIPLE

The system is required to look for reasons it is wrong before it asks a human to accept why it might be right.

10. Time, Retraction, and the Evidence Ledger

PRAMANA treats time as more than a timestamp attached to a document. A fact can be valid during one interval, observed later, believed later still, and revised after a new source appears. This requires temporal state in the data model.

10.1 Event time, observation time, belief time, revision

- Event time: when the represented activity or fact occurred.
- Observation time: when the system captured or observed it.
- Belief time: when the system adopted the resulting claim.
- Revision time: when the belief was superseded or corrected.

10.2 Why the graph cannot be the system of record

A graph edge is visually simple but evidentially incomplete. An edge cannot inherently show that it came from four families, that two items were copies of the same document, that one family was capped, that a contradiction was subtracted, or that one source was retracted later. In PRAMANA, the ledger retains those facts and the graph is rebuilt as a projection.

10.3 Retraction propagation

source retracted

- > evidence superseded
- > dependent relationship recomputed
- > dependent hypothesis recomputed
- > assessment changed / reopened
- > impact log records every affected conclusion
- > historical record remains intact

This means “what did the system believe last month?” becomes a query against stored state rather than an archaeological exercise.

11. Intelligence Modalities

The nine evidence families are complementary. No single modality is assumed to be reliable enough to dominate the entire decision. The value comes from disciplined combination across families.

Family	Examples	Core caution
Cryptographic / PGP	Fingerprint reuse plus metadata such as key creation characteristics, algorithm/size, cipher preferences, subkey structure, UID conventions, and tool artefacts.	Strong but must account for sharing, theft and resale.
Financial	Address reuse, deposit overlap, transaction timing, clustering heuristics.	Never treated as an automatic identity join; mixing and service infrastructure can confound.
Infrastructure	TLS/SAN, server banners, headers, static assets, favicon hashes, analytics IDs, clock skew, host/ASN.	Shared hosting and CDN reuse require rarity and counter-checks.
Contact	Email, Jabber/XMPP, Session, Telegram, Tox and signed migration notices.	Often useful because identifiers can persist across rebrands; unsigned claims remain forgeable.
Content artefacts	Images, EXIF, template structure, prices, units, signature blocks, copied non-boilerplate text.	Useful under prose laundering because operators often copy structure even when wording changes.
Linguistic	Stylometry, character n-grams, style embeddings.	Corroborative only; hard-capped and gated.
Behavioural / temporal	Migration windows, lifecycle co-occurrence, catalogue continuity,	Focuses on event-driven continuity rather than naive timezone inference.

Family	Examples	Core caution
	activity patterns.	
Social / trust	Vouching, reciprocal links, co-membership, moderator statements.	Potentially manufactured; therefore capped.
External corroboration	Authorised legal returns, official notices, credible published corroboration.	High potential value with source-grade and provenance checks.

A particularly important design choice: inverse stylometry

Instead of only asking “Do these two accounts write similarly?”, the system can look for a sudden internal style discontinuity. An abrupt change in style, template, contact pattern, or trust relationships may support the hypothesis that an account was resold, compromised, or moved between staff. The design therefore uses stylometry as both a weak corroborator and a change detector.

PGP metadata as a non-prose signal

PGP key metadata is valuable because it is not simply text. Even when prose is translated or rewritten, operational fingerprints in key construction and tooling may persist. The project identifies this as one of the stronger differentiators beyond simple key fingerprint reuse.

12. AI/ML Architecture and Copilot Governance

The design uses AI/ML selectively. A critical project rule is that generative AI is never authoritative for the attribution decision path.

Layer	Techniques	Role	Authority
Deterministic extraction	Regex, checksum validators, PGP parsing, EXIF, HTTP/TLS, template parsing	Authoritative extraction	Primary
Classical ML	Gradient-boosted rankers, logistic/isotonic calibration, SimHash/pHash/MinHash, TF-IDF rarity	Candidate generation and calibration	Advisory for ranking; authoritative for declared calibration map
NLP	Multilingual NER, language/script ID, transliteration	Normalization and extraction assistance	Validator-gated
Embeddings	Multilingual sentence embeddings, style embeddings, image embeddings	Retrieval and weak corroboration	Advisory and capped
Graph analysis	Recursive CTEs, weighted paths, neighbour similarity	Structure and explanation	Advisory
Statistics	Rarity, Cllr, ECE, bootstrap intervals, temporal splits	Measurement and validation	Authoritative for evaluation outputs

Layer	Techniques	Role	Authority
LLM copilot	Evidence-grounded Q&A, query expansion, drafting assistance	Read-only investigation support	Never authoritative for evidence or attribution
Human analyst	Hypothesis acceptance, conclusion authorship	Final investigative decision	Sole authority

LLM hard rules

- LLM output never writes directly to evidence, pair scores, assessments, must-not-link constraints, or persona records.
- LLM extraction suggestions enter a staging area and must pass deterministic validation before promotion.
- Every copilot answer must cite evidence IDs; uncited answers are not rendered.
- Inference must be visibly distinct from evidence.
- The copilot has no write path to core tables and every interaction is logged with its retrieved evidence set.

PLAIN-ENGLISH INTERPRETATION

The AI can help the analyst read the ledger. It is not allowed to become the ledger.

13. System Architecture and Technology Stack

For the prototype, the project intentionally chooses a compact architecture rather than a fashionable distributed stack. The rationale is reproducibility, offline operation, low operational complexity, and one transactional system of record.



Layer	Prototype choice	Reason
Frontend	React + TypeScript + Vite + Tailwind +	Fast interactive investigator UI with

Layer	Prototype choice	Reason
	TanStack Query + Cytoscape.js	constrained graph expansion.
Backend	Python 3.12 + FastAPI + Pydantic	Same language as extraction and ML layers; avoids duplicated models and extra service boundaries.
Primary store	PostgreSQL 16	Relational, temporal, graph traversal, lexical search, vectors, and queue can coexist at prototype scale.
Graph	Recursive CTEs and materialized views	Keeps graph as projection rather than second source of truth.
Search	Postgres FTS + pgvector + RRF	Hybrid lexical/semantic search without another cluster.
ML/NLP	scikit-learn, sentence-transformers, Splink, imagehash, simhash, changepoint tooling	CPU-viable, explainable, and appropriate to the stated prototype scale.
Queue	Postgres-backed job table with row locking	Avoids Kafka/Redis overhead while retaining restartable jobs.

Why not Neo4j + Elasticsearch + Kafka + Qdrant?

Those systems can each be excellent at their own task. The project rejects them for the prototype because they would create multiple operational surfaces and multiple consistency boundaries before the corpus requires them. The migration path is not denied; it is triggered by measured scale thresholds rather than architecture fashion.

Stated migration triggers

Subsystem	Trigger specified in project
Graph engine	About 5 million edges or 800 ms p95 traversal
Search cluster	About 5 million documents or 400 ms p95 search
Message bus	About 200 documents/sec or more than three consumer groups
Object storage	About 2 TB

14. Analyst Workflow and Product Experience

The interface is organized around investigative questions rather than decorative dashboards. The user should be able to search, inspect a persona, understand a relationship, challenge it, look backward in time, and export a signed case bundle without losing provenance.

Screen / interaction	Purpose
Investigation Workspace	Four synchronized panes for search, evidence, entity context, and active case state.

Screen / interaction	Purpose
Entity Profile	Dossier for accounts, personas, keys, wallets, services, timelines, and review state.
Evidence Balance Sheet	Explains exactly why two identities are linked, including rarity, independence discounts, counter-evidence, and provenance.
Graph Explorer	Degree-limited expansion, layer filters, hub highlighting, strongest-path explanation.
Timeline	Activity, lifecycle shocks, migration windows, and belief-time scrubber.
Persona Comparison	Side-by-side comparison of templates, images, prices, contacts, style, and sessions.
Infrastructure Intelligence	Onion-to-clearnet correlation candidates with rarity and alternate explanations.
Blockchain Intelligence	Address/cluster views with method, mixing flags, provenance, and confidence.
Assessment & Review	Sequential-unmasking gate and analyst decision/rationale.
Reports	Footnoted report generation, signatures, manifests, and export bundles.
Evaluation	Cllr, Tippet, reliability and ablation views as first-class product surfaces.

The three signature interactions

11. Why are these linked? - opens the Evidence Balance Sheet and makes the independence discount visible.
12. Challenge this hypothesis - runs counter-evidence detectors and shows what fired, what did not, and what could change the result.
13. What changed? - scrubs belief time and shows the impact of retractions on downstream assessments.

Sequential unmasking

The review sequence is designed to reduce automation bias: first the analyst sees raw evidence, then the strongest counter-evidence, and only then the computed score. The analyst cannot simply skip to a high-confidence label. The decision requires a written rationale, and machine text remains visually distinct from analyst text.

15. Evaluation Science and Test Harness

The project treats measurement as part of the product, not an appendix. The aim is to determine not merely whether the system ranks examples, but whether its confidence statements are calibrated and whether its central mechanism - independence accounting - changes the behavior in the expected direction.

Evaluation area	Metrics / checks	What the metric answers
Entity resolution	Precision, recall, F1, B-cubed, Variation of Information, false-merge rate, giant-component checks	Are accounts clustered correctly without catastrophic merging?
LR quality	Cllr, Cllr_min, Tippett plots, reliability diagram, ECE, Brier	Do confidence outputs discriminate and calibrate?
Independence accounting	With-vs-without grouping ablation on mirror-heavy cases	Does grouping reduce artificial inflation?
Stylometry	AUC/c@1 and degradation under paraphrase/translation/imitation	How fragile is F6 and how much does the gate help?
Infrastructure	Recovery of planted onion-to-clearnet mappings; false-link rate on clean controls	Does the infrastructure module distinguish rare signals from shared hubs?
Search	Recall@k, MRR, nDCG@10	Can analysts find the relevant evidence?
Graph explanation	Strongest-path correctness against planted paths	Does graph structure explain rather than fabricate?
Human factors	Time-to-first-correct-lead, decision accuracy, over-reliance rate	Does the interface improve or distort analyst judgement?
System performance	Ingest throughput, end-to-end latency, search p95, graph p95, recompute time	Does the prototype meet stated operational budgets?

Temporal and open-set evaluation

Random train/test splits are banned because they can leak time. The planned protocol trains/tunes on earlier months and tests on later months, includes operators unseen during training, samples negatives under realistic class imbalance, and attaches bootstrap confidence intervals to reported numbers.

Why Cllr and Tippett plots matter

Cllr evaluates the cost of calibrated log-likelihood-ratio outputs; a system that merely “guesses” can have poor calibration even when some pairwise ranking metrics look impressive. Tippett plots show the separation behavior of same-operator versus different-operator evidence distributions. Together with reliability diagrams, they test whether the numerical output means what the system says it means.

MEASUREMENT DISCIPLINE

The project record explicitly distinguishes an evaluation target from a measured result. The false-merge target of $\leq 2\%$ is a target, not an achieved accuracy claim.

16. Controlled Test Environments

16.1 RANGE-SIM

RANGE-SIM is a seeded synthetic multi-platform investigation environment with hidden ground truth. Its purpose is not to simulate the entire dark web perfectly; it is to create controlled cases where the team knows the correct answer and can intentionally inject duplicate evidence, hubs, framing attacks, account handovers, LLM laundering, clones, and temporal noise.

- Multiple synthetic marketplaces and forums.
- Planted multi-account operators with different evidence profiles.
- Decoy pairs sharing strong but misleading single-family signals.
- Mirrors, reposts, multilingual and transliterated content.
- Account resale, compromise, and staff-rotation cases.
- LLM-laundered actors whose non-prose artefacts persist.
- Clone marketplace with substituted payment addresses.
- A platform that is later revealed as LE-operated to test retraction propagation.

16.2 RANGE-TOR

RANGE-TOR is a controlled Tor testbed built from services the team owns. The services are intentionally configured with known classes of signals such as exposed status endpoints, controlled TLS/SAN relationships, asset reuse, header leaks, clock skew, and clone/address substitution. The clean control is there specifically to measure false positives.

16.3 Ground-truth isolation

The answer key is kept in a separate schema with no foreign key to operational tables. Continuous-integration tests are intended to fail if the evaluation path can reach it. This prevents leakage from turning evaluation into a disguised lookup.

WHAT SYNTHETIC RESULTS MEAN

Synthetic metrics can demonstrate that the specified mechanisms behave correctly under known cases and controlled noise. They do not prove performance on the live criminal ecosystem.

17. Threat Model, Adversarial Resilience, and Failure Modes

The adversary is assumed to know that correlation systems exist and may deliberately manipulate the traces they leave behind. This is why the project treats evidence generation as an adversarial environment rather than a neutral database problem.

Attack / failure mode	PRAMANA defence	Residual risk
Mirroring / duplicate flooding	Canonicalisation + independence grouping	A novel coordination pattern not modeled by the generator may still fool the grouping.
Hub creation	Rarity weighting + hub suppression + graph degree controls	A slowly growing hub below threshold can still be informative only

Attack / failure mode	PRAMANA defence	Residual risk
		after careful context.
Planted identifiers	Counter-evidence + provenance + cross-family requirement	A well-crafted plant that survives multiple families remains difficult.
Account compromise	Style/template/contact changepoint detection; temporal segmentation	Subtle or long-running compromise can be ambiguous.
LLM / translation laundering	Stylometry gating + non-prose artefacts	If all modalities are laundered, correlation may collapse.
Shared infrastructure	Rarity weighting + shared-host counter-check	Rare shared infrastructure can still create false candidates.
Transitive graph poisoning	No automatic transitive closure + must-not-link constraints + component caps	Complex graphs remain computationally and cognitively difficult.
Compartmented operator with no shared traces	None; correlation has no input	A disciplined adversary can be invisible to any correlation system.

The most important residual risk

A disciplined actor who shares nothing across identities is fundamentally difficult to identify with a correlation system. The project explicitly acknowledges that no architecture can infer evidence that does not exist. This is not a bug to hide; it is a boundary condition of the problem.

Why adversarial evaluation is central

The project plans to make attack degradation a headline result. An architecture that looks strong only on clean, friendly data is not enough for this problem. The test harness therefore includes both planted traps and ablation comparisons.

18. Legal, Forensic, Ethical, and Governance Posture

The project is explicitly framed for an Indian context and adopts a set of legal/forensic design considerations. The important distinction is between “designed to support good practice” and “legally guaranteed admissible.” PRAMANA makes the former claim, not the latter.

Instrument / practice named in project	Engineering implication
Bharatiya Sakshya Adhiniyam 2023, Section 63	Export includes a certification-oriented integrity workflow; the project does not claim automatic admissibility.
DPDP Act 2023	Data minimisation, purpose logging, and no assumption of State-instrumentality exemption for a student team.
CERT-In directions 2022	Consideration of incident/log handling obligations in authorised deployments.
ISO/IEC 27037	Identification, collection, and preservation disciplines for digital evidence.
ENFSI evaluative reporting guidance	Likelihood-ratio framing and the need for empirical

Instrument / practice named in project	Engineering implication
	validation; the project does not claim ENFSI validation is already complete.
Sequential unmasking	Reduce analyst bias by controlling when evidence, counter-evidence, and scores become visible.
WARC/WACZ + hashes + RFC 3161	Provenance and integrity support from capture through reporting.

Important corrections in the project research audit

- The project rejected an overbroad claim that AI paraphrasing had definitively broken stylometric verification; the cited studies were not evidence for that exact inference.
- The project rejected an “85%” historical attribution statistic because its source was a five-case hand-coded sample with selection bias.
- The project rejected the statement that Section 63 makes missing immediate hash chains automatically inadmissible; it instead treats capture-time sealing as a strong engineering best practice aligned with evidence integrity.
- The project rejected the use of Dempster-Shafer / Subjective Logic as the principal fix for non-independence and replaced it with explicit independence grouping and capped log-LR fusion.

GOVERNANCE RULE

Do not turn a design safeguard into a legal absolute. The product can create strong preconditions for defensibility; admissibility and lawful identity attribution remain matters of authorised process and competent decision-makers.

19. Implementation Status and Roadmap

The supplied project record is unusually explicit about status. That should be preserved in any presentation or submission.

Phase	Work	Status
1	Research instrument	Complete
2	Research pass and source re-grading	Complete; findings audited and selectively rejected/adopted
3	Master blueprint, verification checklist, research integrity audit	Complete
4a	Presentation specification	Complete
4b	Presentation generator / deck build specification	Complete as a deliverable; execution output is separate
5	V1 implementation against RANGE-SIM	Not started in the project record

Phase	Work	Status
6+	Remaining implementation, evaluation, rehearsal and deployment path	Specified, not yet evidenced as completed

Implementation priorities

14. Build the evidence ledger and provenance model first; every downstream feature depends on this.
15. Implement deterministic extractors with adversarial unit tests and canonicalisation regressions.
16. Implement entity resolution with hard constraints and giant-component protections.
17. Implement fusion, counter-evidence, caps, calibration, and replayable traces.
18. Build search/graph query and the Evidence Balance Sheet before cosmetic dashboard work.
19. Build the evaluation harness early enough that the UI cannot consume the schedule at the expense of scientific measurement.
20. Run the full ablation and adversarial suite, then freeze parameters and rehearse an offline deterministic demonstration.

Why the evaluation harness is not optional

The final red-team section of the project explicitly identifies a major schedule risk: the temptation to sacrifice evaluation to finish the UI. The project itself says this is the wrong trade because the evaluation harness is what turns the central design argument into a measurable claim.

20. Strategic Assessment

20.1 What is genuinely strong

- The project identifies a concrete and defensible failure mode: correlated evidence being treated as independent evidence.
- The evidence ledger is a better system-of-record abstraction for reversible investigative reasoning than a graph alone.
- The design uses known, interpretable foundations - probabilistic linkage, rarity weighting, likelihood ratios, temporal data modeling - rather than inventing an opaque new score for novelty.
- The interface is built around explanatory interactions rather than vanity metrics.
- The project is unusually disciplined about saying what is not yet measured.

20.2 What remains scientifically open

- The independence-grouping mechanism is still a design hypothesis until its ablation is run.
- Evidence-family caps are judgement-based initial priors and require sensitivity and later calibration.
- Absolute population-level likelihood ratios are difficult because the true global reference population is unknown.
- Synthetic testbeds cannot guarantee transfer to real adversarial behaviour.
- A disciplined actor who deliberately shares no observable traces may remain invisible.

20.3 Why the composition matters

The project's strongest innovation claim is architectural composition, not a single novel model. Canonicalisation, independence keys, rarity, caps, counter-evidence, must-not-link semantics, bitemporal history, provenance, and calibration reinforce one another. Removing any one of them reintroduces a specific failure mode that the others cannot fully repair.

FINAL THESIS

PRAMANA turns fragmented observations into a contestable evidential statement. It does not promise certainty; it makes uncertainty structured, visible, reversible, and accountable.

Appendix A. One-Page Technical Summary

Question	Answer
What is PRAMANA?	An evidence-centric attribution platform for dark-web threat actors.
Core problem?	Evidence arithmetic under duplication, correlation, rarity shifts, contradiction, adversarial manipulation, and temporal revision.
Central idea?	Weight of evidence, not volume of evidence.
Three pre-score tests?	Independence, rarity, exclusion.
Core score?	Capped, damped log10 likelihood-ratio fusion.
Global ceiling?	4.0 log10 LR.
Minimum independent families?	$k \geq 2$.
Hard contradiction?	EXCLUDED, no numeric score.
Graph role?	Projection / view; ledger is source of truth.
Identity merge rule?	Probabilistic linkage + hard constraints + capped clustering; no naive transitive closure.
Main demo?	Evidence Balance Sheet with visible independence discount.
AI role?	Assistance and retrieval; no authority over evidence or final attribution.
Prototype store?	PostgreSQL 16.
Evaluation?	Cllr, Cllr_min, Tippett, reliability/ECE, clustering metrics, ablations, adversarial degradation.
Current status?	Designed/specified and presentation-ready; implementation and benchmark results are not yet claimed.

The mental model

TRACE -> EVIDENCE -> RELATIONSHIP -> HYPOTHESIS -> COUNTER-EVIDENCE
-> FUSION -> ASSESSMENT -> HUMAN DECISION -> REPORT

Appendix B. Glossary

Term	Definition
Attribution	An evidential claim about whether identified pseudonymous surfaces are controlled by the same underlying actor, stopping short of automatic real-world identity.
B-cubed	A cluster-level metric that evaluates entity-resolution

Term	Definition
	precision and recall.
Bitemporal	A model that preserves more than one concept of time, especially event validity and system belief/revision time.
Canonicalisation	Deduplicating materially identical observations into a canonical artefact representation.
Cllr	A cost-based metric for evaluating likelihood-ratio calibration and discrimination.
Defence hypothesis	The competing proposition against which the same-actor proposition is evaluated, usually that the identities are controlled by different actors.
Evidence family	One of nine broad signal classes with its own contribution ceiling and reliability assumptions.
Evidence group	A set of within-family observations treated as sharing an independence origin.
EXCLUDED	A hard state meaning a must-not-link condition fired; it is intentionally not represented as a numeric confidence score.
Fellegi-Sunter	A classical probabilistic record-linkage framework for comparing match and non-match evidence.
Hub	A high-frequency indicator that connects many records and therefore carries low identity information.
IDF-like rarity	A weighting idea in which an indicator is more informative when it is uncommon in the reference population.
Independence key	A field used to group evidence that shares a common underlying observation source.
Likelihood ratio	The ratio of how probable the observed evidence is under one proposition versus a competing proposition.
log10 LR	Base-10 logarithm of a likelihood ratio; additive across properly handled evidence contributions.
Must-not-link	A constraint that forbids a particular merge regardless of additional weak support.
pHash	Perceptual image hash designed to identify visually similar images despite small transformations.
Sequential unmasking	A review procedure that controls when the analyst sees raw evidence, counter-evidence, and computed score.
Tippett plot	A plot used to inspect separation of likelihood-ratio behavior for same-source and different-source trials.
Transitive closure	The graph shortcut that can infer $A=C$ from $A=B$ and $B=C$; PRAMANA explicitly prevents this from becoming an automatic identity conclusion.

Appendix C. Judge / Viva Questions and Defensible Answers

Is this just a dark-web scraper?

No. Collection is an input. The product is the evidence ledger and the accounting layer that decides what each trace is worth.

What is actually innovative?

Making evidence independence a first-class data property and making every assessment explainable, reversible, and measurable rather than hiding everything inside an aggregate similarity score.

Why not Dempster-Shafer / Subjective Logic?

The project audit rejected it as the wrong principal remedy for duplicated/correlated evidence and replaced it with explicit independence grouping and capped log-LR fusion.

Why not simply use an LLM?

Because a generative model cannot be allowed to silently invent, merge, or strengthen evidence. The LLM is restricted to read-only evidence-grounded assistance.

Why not Neo4j?

The prototype does not need a second authoritative graph store. PostgreSQL can provide recursive traversal while keeping evidence and relationships transactional.

How do you stop a giant component?

Hub suppression, removal of hub edges before closure, must-not-link constraints, component-size caps, and no automatic transitive closure.

How do you handle duplicated evidence?

Canonicalise first, then group by independence key. Within a family only the strongest item in a group counts; later independent groups are damped.

How do you stop one strong modality from dominating?

Every evidence family has a contribution cap and a conclusion requires at least two independent families.

Why is stylometry capped?

Because it is vulnerable to topic confounds, translation, imitation, and LLM mediation. It is useful as corroboration, not as a solitary identity oracle.

Can you identify a real person?

Not automatically. The highest in-system output is a reviewed attribution hypothesis at pseudonym level. Real-world identification requires lawful external process.

Are the $\leq 2\%$ false-merge numbers real?

No. That is a design target. The project record explicitly says the system has not yet been benchmarked.

Can this de-anonymize Tor?

No. The design rejects network-wide enumeration and instead uses lawful seeded collection and analysis of observable operator mistakes or infrastructure relationships.

What happens when evidence is retracted?

The ledger preserves the old state, supersedes the retracted input, recomputes affected relationships and assessments, and records the impact.

Why use synthetic data?

Because it provides hidden ground truth and allows controlled adversarial traps and ablations that public real-world datasets cannot provide.

What is the biggest unsolved limitation?

A disciplined operator who deliberately creates no shared observable traces may remain invisible to any correlation system.

Appendix D. Claim Discipline Checklist

Use this checklist whenever the project is presented to a judge, faculty panel, reviewer, or technical stakeholder.

Say this	Do not say this
“Design target $\leq 2\%$ false-merge rate.”	“We achieve 2% false merges.”
“The system produces calibrated investigative leads once evaluated and calibrated.”	“The system proves who the criminal is.”
“The graph is a projection of the evidence ledger.”	“The graph itself is the ground truth.”
“LLM assistance is read-only and evidence-grounded.”	“Our AI autonomously attributes actors.”
“Stylometry is capped and corroborative.”	“Stylometry identifies the author with certainty.”
“The design supports forensic/traceability practices.”	“The software guarantees court admissibility.”
“Synthetic evaluation measures mechanism behavior under controlled conditions.”	“Synthetic accuracy proves live-world performance.”
“The current project record has implementation as the next phase.”	“The prototype has already been deployed in production.”

ONE SENTENCE TO CLOSE A PRESENTATION

PRAMANA does not promise certainty. It makes every attribution lead show its evidence, its discounts, its contradictions, its provenance, and its limits - so a human can decide what deserves further lawful investigation.